Verity

Intelligent brand governance. Every word. Every pixel. On brand. · Team Merge Conflicts · Inkloom AI Hackathon, Brand Governance track · Round 3 final project document

Working product (no signup): https://verity-two-delta.vercel.app → click **Start judge walkthrough**. Roles: Maya (Creator), Rajan (Manager), Anita (Approver), one click each on the landing page. Maya also belongs to a second brand, Northwind, where she is a manager.
Source code (public): https://github.com/Swas00/verity · **API reference:** `docs/openapi.yaml` · **Round 3 results page in the app:** `/roadmap`
Note: the site runs on free hosting. After a quiet period the first visit can take 30 to 60 seconds; the landing page says "Waking up the server" and signs you in by itself.

1. What Verity does, and what Round 3 added

Verity checks marketing content against a brand's guidelines and explains every flag with the clause it breaks, a confidence score and a suggested fix. People make the final call: Creator, Manager and Approver roles with a tamper-evident audit trail. Nothing is published automatically. Round 2 delivered the text workflow (guideline builder, TermGuard, ClaimGuard, ToneGuard in basic mode, MetaAgent scoring, approval workflow, audit trail, dashboard). Round 3 adds the three improvements promised, and closes several Round 2 gaps.

2. The three promised improvements: built, tested, measured

Improvement	What a user can do now	Target and measured result
VisualGuard images and logos	Upload a banner or graphic. Every problem is marked with a numbered box and its guideline clause: colours outside the palette, a retired logo, a stretched, squashed, recoloured or crowded approved logo. Managers keep each brand's logos, retired logos and palette on the Brand page. Decisions, scoring, approval and audit work as for text.	**Target:** 85% recall, 10% false flags, 30 images. **Held-out:** 22 of 22 found, 0 of 8 compliant flagged. **Stress set** (harder, never tuned): 9 of 15; rotated and cropped logos missed, photo backgrounds flagged as off-palette. Images are synthetic.
Multi-brand workspaces agency mode	One person, several brands, a different role in each (Maya: creator at Lumina, manager at Northwind). A header switcher changes guidelines, reviews, audit, dashboard, drift and notifications. Lumina wording in a Northwind post is flagged "other brand". Managers send clients a read-only report link that expires and can be revoked.	**Target:** no cross-brand access; 80% of planted cross-brand wording flagged. **Result:** every review route called from the other brand returns 404 (the test fails if the filter is removed); wording: 18 of 21 (86%), 0 of 18 clean texts flagged. Voice-only wording is not caught (0 of 3).
Brand Drift Intelligence	The Drift page shows which teams are moving away from the guidelines: recent violation rate against the team's own baseline, the clauses most likely to be broken next, trend charts with data tables, a weekly digest and a manager alert.	**Target:** alert within 3 weeks of the drift starting; none on a stable team. **500 held-out simulated series:** stable teams alerted 4.6%; 2.2x step caught within 3 weeks in 92% (median 1 week); ramp 71%; mild

		1.6x step only 45%. Seeded history: Social found, three teams stable.

Round 2 gaps closed: ClaimGuard no longer flags "Limited time only" or "best-in-class" as claims; uploads up to the documented 5 MB now work (the API limited every request to 2 MB); the queue loads 25 reviews at a time with title search; results can be shared with a client.

3. Finished and working on the live link

Area	What works
Guidelines	Upload PDF, DOCX, TXT or MD (or a sample); editable preview with notes; publish as an immutable version. Per-brand logos, retired logos and palette.
Text review	Editor with highlights on every flagged span; TermGuard (variants, allowed contexts), ClaimGuard (claims register with document and page, number conflicts), ToneGuard in labelled basic mode; MetaAgent score; results stream in; one failing agent never blocks the others; accept, edit, dispute (reason required); revision diff.
Image review	Upload; numbered boxes on the image; the same problems as a text list; dispute or override with a reason; upload new versions; submit and approve.
Workflow and audit	Submit, approve, reject, escalate, final approval, send back, publish; server-enforced roles; no self-approval; queue; comments; notifications; append-only hash-chained audit trail per brand with verification and CSV or JSON export.
Brands	Switcher; per-brand roles; other-brand wording flagged; client report links (hash-stored token, expiry, revocation, no content or names).
Insight	Dashboard (scores, time to approve, violations by agent, content type and creator, clauses reviewers keep overriding, learning loop); Drift page and weekly digest.
Engineering	OpenAPI reference for every route (checked against the code by a test); pagination and search; request ids, logging and security headers; validation; optional Redis queue; server-waking message; accessibility checks and keyboard-only runs in light and dark mode.

4. Architecture and technology

Service	Technology and role
Web (Vercel)	Next.js 14, React 18, TipTap. Interface only; every permission is re-checked by the API.
API (Render, free)	Node.js and Express. JWT demo logins; a brand-context middleware that sets the active brand and the user's role in it; workflow state machine; SSE; cache and rate limiter; audit log; drift analysis; client reports.
AI service	Python FastAPI, Pydantic, Pillow, NumPy, OpenCV (headless). Text agents, MetaAgent, VisualGuard. Stateless: guidelines, logos and palette are sent with each request. Same container as the API.
Database (Neon, free)	PostgreSQL with JSONB; memberships, assets and images (bytea), share links, drift digests; `brand_id` on every table.

Hard parts

- **Logo checks without a model.** Template matching on edge maps: a coarse search over 22 sizes and 11 stretches, then a precise match; a stretched fit must score clearly higher than the undistorted logo to

count as distortion; a second pass on colour edges finds a recoloured logo; recolouring is judged per logo colour.

- **Provable brand isolation.** One middleware decides the brand and role; a test calls every review route from the other brand; breaking the filter on purpose makes that test fail.
- **A drift detector that stays quiet.** A Poisson standard error widened by a measured dispersion factor, a minimum rise of 1.3 times, and a threshold chosen on one simulated set and measured on another.
- **Honest degradation.** Per-agent isolation, labelled fallbacks, a circuit breaker around the optional queue, memory kept under the free host's limit (about 210 MB peak for a large image).

5. Testing notes

- **Automated:** 102 API tests run by default (4 more need Redis or the real AI service, and run in CI), 144 AI tests, 7 web unit tests, lint, build, and browser suites in real Chrome: full journey, accessibility audit (0 serious or critical violations), keyboard-only runs, 320 px reflow, sleeping server, multi-brand, drift, image reviews against the real AI service, queue, Round 3 keyboard use. All run in GitHub Actions on every push.
- **Evaluation sets:** text agents (30 + 30 texts), contamination (39 texts), images (30 dev + 30 held-out + 20 stress), drift (simulated teams). Thresholds were set on dev data. Held-out and stress sets were not used to tune, except that the two ClaimGuard patterns fixed in Round 3 were already-known held-out errors.
- **Found by the new tests while building:** stale rows after switching brand; weeks starting on Thursday; images returned as JSON; a text charset added to image types; the 2 MB request limit; plain banners matched as a retired logo.

6. What changed from the plan, and why

- VisualGuard uses classical image processing, not CLIP: CLIP needs more memory than the free 512 MB host and a model download. It costs nothing and takes 1 to 2 seconds per image.
- Drift uses violation rates, not tone embeddings (no embedding model on the free host; the embedding experiment in Round 2 did not help).
- No S3: logos and reviewed images live in PostgreSQL. The weekly digest is created on first view of a new week because the free host has no scheduler. Teams are a field on a brand membership.

7. What does NOT work yet

- ToneGuard on the demo is rule-based basic mode, so subtle voice problems are not caught.
- VisualGuard misses rotated and cropped logos and treats photographs as off-palette; it knows the brand's logo files, not "off-brand" in general. Test images are synthetic, so real designs will score lower.
- Other-brand detection does not catch wording with no listed term, claim or name. Independent test data written by a teammate showed ClaimGuard missing health, comparison and duration claims (recall 0.17); it was fixed afterwards, so the improved figure (0.86 / 1.00) is not independent and needs a fresh set. ToneGuard's basic mode still misses idiomatic hype.
- Drift needs about 7 weeks of history per team; mild drifts are often missed; the history is simulated and seeded.
- No usability test with outside people and no screen-reader test. The demo data is shared by every visitor. The free host sleeps when idle. Email is built but off.

8. Team and disclosures

Name	Round 1 role	Contribution (as the repository shows it)
Swastik Sahu	Full-Stack Lead and AI/ML Engineer	Project lead and integration; Round 3 design and build of all three improvements, tests, evaluation, documentation and CI.
Akriti Tigga	Backend Engineer	Ran the whole API test suite on a real PostgreSQL 18.4 (102 passed, 0 failed), started the server on it and checked the Round 3 routes, the seeds and the audit chain, wrote a security review of the share links, uploads and brand isolation (docs/SECURITY_REVIEW.md), found that share-link tokens were being written to the request log, and fixed it with a regression test (PR #3).
Gayathri VS	Frontend Engineer and UI/UX Designer	Round 2: accessible reason dialog with end-to-end test updates.
Sakshi Kumari	AI / ML Engineer	Wrote an independent test set for the agents: 40 cross-brand texts (ai/eval/contamination_independent.json) and 30 fresh texts for the term, claim and tone agents (ai/eval/heldout2.json), and made the two evaluation scripts accept a named set (PR #4). Her results showed that ClaimGuard missed health, comparison and duration claims (recall 0.17), which was then fixed. The real-image test set that was also requested has not been delivered.
Shivam Agarwal	Backend Engineer and DevOps	Round 2: optional BullMQ and Redis queue, queue tests and benchmark, CI jobs with and without Redis.

AI assistance: the team used Claude (Anthropic) while building, including Claude Code for writing and testing parts of the code, tests and documentation; the team directed and reviewed the work. **Licence:** MIT. **Demo data:** "Lumina Coffee Co.", "Northwind Tea Co." and every review, user, team and image are fictional.